PROJECT CASE STUDY

Air Pollution Prediction & Anomaly Detection

Regression Benchmark + Isolation Forest

Category	Status	Year
Applied ML	Completed	2025

An end-to-end notebook comparing six regression models for AQI prediction and adding unsupervised anomaly detection for unusual pollutant combinations.

23,463	0.9989	1,174
source rows	best recorded R2	anomalies

Context and problem

Global air-quality records require cleaning, comparison across multiple regressors, and a way to surface unusual observations beyond standard prediction metrics.

Strategy and implementation

- Performed EDA, missing-value handling, outlier inspection, and feature preparation.
- Benchmarked Linear Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost, and SVR.
- Used cross-validation alongside hold-out metrics.
- Applied Isolation Forest to detect anomalous pollutant profiles.

My role

Complete notebook implementation, model benchmarking, anomaly analysis, output export, and result interpretation.

Technology stack

Python | Pandas | scikit-learn | XGBoost | Isolation Forest | Matplotlib

Results and evidence

- Processed 23,463 source records.
- Best recorded Gradient Boosting result: R2 0.9989 and RMSE 1.8947.
- Flagged 1,174 rows as anomalies, approximately 5% of evaluated records.

Key takeaways

- Very high AQI scores require leakage-aware interpretation because pollutant AQI components are closely related to the target.
- Benchmark tables should include speed and cross-validation, not only the best hold-out score.

Portfolio: github.com/awai1s | awaisraza5424@gmail.com